

SUMMARY

ML & Robotics Engineer with 3+ years of industry experience in ADAS testing, vehicle system integration & data analysis. Currently completing MS in Electrical Engineering with a focus on deep reinforcement learning, POMDPs, and diffusion models for autonomous systems. Proficient in Python, ROS2, Julia, and modern ML frameworks (TensorFlow, PyTorch).

EDUCATION

University of Colorado - Boulder GPA - 3.96 <i>Key Courses: Advanced Robotics, Transformer-based Robotic AI, Decision Making under Uncertainty, Convex Optimization, Machine Learning, Deep Learning, Linear Control Design, Computer Vision</i>	MS, Electrical Engineering (Focus on Robotics & AI) <i>Expected Graduation – May 2025</i>
University of Michigan GPA - 3.57	BSE, Mechanical Engineering <i>Graduation – May 2020</i>

WORK EXPERIENCE

APTIV Lead Vehicle Systems Test Engineer - Led design & execution of 700+ vehicle-level system tests across multiple vehicle platforms; supervised a team of 9 engineers. - Reduced testing cycle time by 40% through streamlining data analysis process, and introducing shorter iterations for test cycles. - Standardized documentation and data analysis for 22 engineers across 3 teams, cutting data analysis time by 30%. Vehicle Systems Test Engineer - Performed 50+ hours/week of on-road & closed-course vehicle testing on ADAS/AD prototypes, using CAN loggers, ethernet monitoring, proprietary validation tools, and custom MATLAB scripts. - Led autonomous vehicle testing initiatives, from route design to data acquisition and root-cause analysis. Vehicle Systems Integration Engineer - Integrated ADAS/AD systems at bench and vehicle levels, employing Vector CANoe, CANape, vFlash, and Lauterbach to ensure effective inter-module communication and system functionality. - Collaborated daily with Systems, Software, and Algorithm teams to resolve test system configurations and streamline feature integration.	Troy, MI <i>Feb 2022 – Jun 2023</i> <i>May 2021 – Jan 2022</i> <i>Sep 2020 – Apr 2021</i>
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PROJECTS & RESEARCH

Advanced Robotics Implemented SLAM and obstacle detection with LIDAR on an AWS DeepRacer using ROS2. Integrated YOLOv3 for stop-sign recognition, tackled system memory management, and debugged network communication issues.	CSCI 5302
Transformer-Based Robotic AI Investigated advanced RL via Deep Transformer Q-Networks, published a blog article on CLIP embeddings, and evaluated cutting edge LLM/VLM models for robotics tasks.	CSCI 7000
Decision Making under Uncertainty Implemented Q-Learning, SARSA, MCTS, & SARSOP for a variety of MDPs/POMDPs.	ASEN/CSCI 5264
Generative Models Working under Dr. Bradley Hayes to explore diffusion-based generative models for improved policies in partially observable domains.	Ongoing Research

SKILLS

Programming & Tools: Python, Julia, MATLAB, TensorFlow, PyTorch, JIRA
Systems: ROS2, Vector CAN tools (vFlash, CANoe, CANape), Polarion, Solidworks
Methodologies: Reinforcement Learning, Systems Engineering, Robotics & Autonomous Systems, Testing & Integration, Motion Planning, Machine Learning, Computer Vision, Optimization